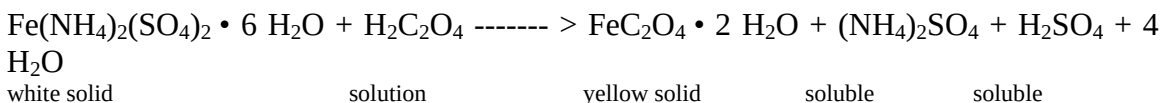


# Experiment 2 - Stoichiometry Experiment

## The Preparation of an Oxalate Compound

### Introduction

In this experiment, you will synthesize potassium ferric oxalate,  $\text{K}_3\text{Fe}(\text{C}_2\text{O}_4)_3 \cdot 3 \text{H}_2\text{O}$  in a two-step process. The first reaction is:



The main product of the first reaction,  $\text{FeC}_2\text{O}_4 \cdot 2 \text{H}_2\text{O}$ , will be separated from the other products, which remain in solution. In the second reaction, the yellow solid,  $\text{FeC}_2\text{O}_4 \cdot 2 \text{H}_2\text{O}$ , will be converted to  $\text{K}_3\text{Fe}(\text{C}_2\text{O}_4)_3 \cdot 3 \text{H}_2\text{O}$ , according to the following reaction:



The names and molecular weights of important compounds in this experiment are listed in the table below.

| Formula                                                                  | Name                                 | Molar Mass (g/mol) |
|--------------------------------------------------------------------------|--------------------------------------|--------------------|
| $\text{Fe}(\text{NH}_4)_2(\text{SO}_4)_2 \cdot 6 \text{H}_2\text{O}$     | ferrous ammonium sulfate hexahydrate | 392                |
| $\text{FeC}_2\text{O}_4 \cdot 2 \text{H}_2\text{O}$                      | ferrous oxalate dihydrate            | 180                |
| $\text{H}_2\text{C}_2\text{O}_4$                                         | oxalic acid                          | 90                 |
| $\text{K}_2\text{C}_2\text{O}_4 \cdot \text{H}_2\text{O}$                | potassium oxalate monohydrate        | 184                |
| $\text{K}_3\text{Fe}(\text{C}_2\text{O}_4)_3 \cdot 3 \text{H}_2\text{O}$ | potassium ferric oxalate trihydrate  | 491                |

The main objectives of this experiment are to prepare as great a yield of  $\text{K}_3\text{Fe}(\text{C}_2\text{O}_4)_3 \cdot 3 \text{H}_2\text{O}$  as possible and to practice the mathematics of stoichiometry. To measure the efficiency of the procedure, you will calculate the percent yield:

$$\text{Percent Yield} = \frac{\text{Actual Yield}}{\text{Theoretical Yield}} \times 100\%$$

The *theoretical yield* is the calculated maximum amount of product that can be obtained using your starting materials. In this experiment, the theoretical yield is the maximum amount of  $\text{K}_3\text{Fe}(\text{C}_2\text{O}_4)_3 \cdot 3 \text{H}_2\text{O}$  that can be produced using the specified amounts of starting materials. Remember to determine the limiting reagent before determining the theoretical yield.

The *actual yield* is the amount of product that you actually obtain (in grams).

In order to calculate the theoretical yield for the reaction, you will need to use both reactions. One of the products from the first reaction,  $\text{FeC}_2\text{O}_4 \cdot 2 \text{H}_2\text{O}$ , is a reactant in the

second reaction. However, although the stoichiometric coefficient of the first reaction for  $\text{FeC}_2\text{O}_4 \cdot 2 \text{H}_2\text{O}$  is 1, the stoichiometric coefficient for the second reaction is 2. You must account for this fact in your limiting reagent and theoretical yield calculations.

### Procedure

*Note – There are MANY color changes in this experiment. Your lab notebook should reflect this. For example, if you add a reagent, and the solution turns purple, write “turned purple” in your lab notebook in the data portion (on the right-hand side). These observations MUST be noted. Furthermore, although the procedure says to add “25 mL” to something, it is important that you record the actual volume added to the nearest 0.1 mL.*

Tare a 200 mL beaker and place approximately 5 g of  $\text{Fe}(\text{NH}_4)_2(\text{SO}_4)_2 \cdot 6 \text{H}_2\text{O}$ . Record mass to the nearest 0.001 g. Dissolve the sample in 15 mL of distilled water to which 5 drops of 3 M  $\text{H}_2\text{SO}_4$  have been added. To this solution, add 25 mL (using graduated cylinder) of 1 M  $\text{H}_2\text{C}_2\text{O}_4$  (oxalic acid) solution. Record the actual volumes to the nearest 0.1 mL.

Set up a ring stand so that the beaker is supported on a ring covered with wire gauze. Place another ring so that it surrounds the top of the beaker to help secure the beaker from falling. Using a Bunsen burner, heat the mixture until boiling, stirring continuously to prevent spattering. DO NOT LEAVE UNATTENDED. If the solution starts to boil over, remove the flame. Allow the yellow precipitate (solid) of  $\text{FeC}_2\text{O}_4 \cdot 2 \text{H}_2\text{O}$  to settle. Using large beaker tongs, carefully decant (pour off) the hot supernatant liquid into a waste beaker, and keep the solid in the 200 mL beaker. Wash the yellow precipitate by adding about 20 mL of distilled water to the beaker. Reheat using the Bunsen burner, stir, and allow the solid to settle. Decant the liquid into the waste beaker, and keep the solid.

To the yellow solid ( $\text{FeC}_2\text{O}_4 \cdot 2 \text{H}_2\text{O}$ ) in the beaker, add 15 mL of saturated (300 g  $\text{K}_2\text{C}_2\text{O}_4 \cdot \text{H}_2\text{O}$  per liter) potassium oxalate ( $\text{K}_2\text{C}_2\text{O}_4 \cdot \text{H}_2\text{O}$ ) solution. Record this volume to the nearest 0.1 mL. Support your thermometer using a ring stand, a one-holed stopper, and a clamp. Place the thermometer in the solution. Carefully heat the solution to 40 °C. Obtain 20 mL of 3%  $\text{H}_2\text{O}_2$  in your 25 mL graduated cylinder. Record this volume to the nearest 0.1 mL. Add the  $\text{H}_2\text{O}_2$  **slowly**, a few drops at a time, stirring continuously and keeping the temperature near 40 °C. Do not worry if you see a slimy red precipitate (iron (III) hydroxide). After all of the  $\text{H}_2\text{O}_2$  has been added, heat to boiling. Once the solution boils, add 8 mL of 1.0 M  $\text{H}_2\text{C}_2\text{O}_4$ . Record this volume to the nearest 0.1 mL. Add the first 5 mL all at once, and the last 3 mL *very slowly*, keeping the solution boiling.

Set up a funnel with filter paper on a ring stand and filter the boiling solution into a clean 100 mL beaker. Allow to cool to room temperature, and then add 10 mL of ethyl alcohol to the beaker. Cover your beaker with your largest beaker, and place it in your laboratory desk until the next laboratory period.

The reason that the beaker should be placed in your desk is that the product,  $\text{K}_3\text{Fe}(\text{C}_2\text{O}_4)_3 \cdot 3 \text{H}_2\text{O}$ , decomposes in the presence of light, producing  $\text{CO}_2$  and an iron (II) compound. The reason to cover the beaker is to prevent complete evaporation of the water, which would make it impossible to obtain pure  $\text{K}_3\text{Fe}(\text{C}_2\text{O}_4)_3 \cdot 3 \text{H}_2\text{O}$  crystals. Furthermore, slow evaporation encourages the development of larger, purer crystals.

### Next lab session

Assemble a filtration set up. Using a stirring rod equipped with a rubber policeman, transfer the crystals of  $\text{K}_3\text{Fe}(\text{C}_2\text{O}_4)_3 \cdot 3 \text{H}_2\text{O}$  from the beaker to the filter paper in the funnel. When the crystals are free of the solution, remove the filter paper and spread the crystals so that they can dry. Find the mass of a dry weighing boat. When the crystals are dry, scrape the crystals from the filter paper into the preweighed boat. Determine the mass of the crystals by subtracting the mass of the empty boat from the mass of the full boat. Record the mass to the nearest 0.001 g. Describe the crystals (color, shape, size, etc.) in your lab notebook. Show the crystals to the instructor before disposing of them in the proper waste container.

### Prelab

In addition to writing up the procedure, answer the following questions in your lab notebook. Show all work.

1. A student reacts 4.991 g of  $\text{Fe}(\text{NH}_4)_2(\text{SO}_4)_2 \cdot 6 \text{H}_2\text{O}$  with 26.2 mL of 1 M  $\text{H}_2\text{C}_2\text{O}_4$  solution to produce the yellow solid,  $\text{FeC}_2\text{O}_4 \cdot 2 \text{H}_2\text{O}$ . This solid is then reacted with 15 mL of saturated  $\text{K}_2\text{C}_2\text{O}_4 \cdot \text{H}_2\text{O}$  solution (300 g  $\text{K}_2\text{C}_2\text{O}_4 \cdot \text{H}_2\text{O}$  per liter of solution), 20 mL 3%  $\text{H}_2\text{O}_2$ , and 8 mL of 1.0 M  $\text{H}_2\text{C}_2\text{O}_4$ . The student follows the procedure in the lab handout.

Determine:

- a. Moles of  $\text{Fe}(\text{NH}_4)_2(\text{SO}_4)_2 \cdot 6 \text{H}_2\text{O}$
- b. Moles of  $\text{H}_2\text{C}_2\text{O}_4$  (initially added)
- c. Moles of  $\text{K}_2\text{C}_2\text{O}_4 \cdot \text{H}_2\text{O}$  (note – need to use the fact that there are 300 g  $\text{K}_2\text{C}_2\text{O}_4 \cdot \text{H}_2\text{O}$  per liter of solution)
- d. Based on your answers from parts a-c, what is the limiting reagent? (Assume all other reagents are in excess.) *(Hint – The best way to do this is to determine which is limiting between  $\text{Fe}(\text{NH}_4)_2(\text{SO}_4)_2 \cdot 6 \text{H}_2\text{O}$  and  $\text{H}_2\text{C}_2\text{O}_4$ . Then, determine what is limiting between that compound and the final compound,  $\text{K}_2\text{C}_2\text{O}_4 \cdot \text{H}_2\text{O}$ .)*
- e. What is the theoretical yield?
- f. If the student obtains 4.881 g of product ( $\text{K}_3\text{Fe}(\text{C}_2\text{O}_4)_3 \cdot 3 \text{H}_2\text{O}$ ), what is the percent yield for the reaction?

### During the lab

Be sure to record actual masses and volumes used. Describe any color changes seen, as well as the point at which the color change occurs during the experiment. The right-hand side of the lab notebook would be a good place to make these observations.

### Postlab

1. Perform the following calculations. Show all work.
  - a. Moles of  $\text{Fe}(\text{NH}_4)_2(\text{SO}_4)_2 \cdot 6 \text{H}_2\text{O}$
  - b. Moles of  $\text{H}_2\text{C}_2\text{O}_4$  (*first addition*)
  - c. Moles of  $\text{K}_2\text{C}_2\text{O}_4 \cdot \text{H}_2\text{O}$
  - d. Moles of  $\text{H}_2\text{C}_2\text{O}_4$  (*second addition*)
  - e. Calculation of limiting reagent:
  - f. Limiting reagent
  - g. Theoretical Yield
  - h. Percent Yield
  
2. Chemists that work in industry often do syntheses in which products from one reaction are carried on to be reactants in the next reaction. In multi-step syntheses, they often try to find the exact reaction in a procedure in which they lose yield. If you could change the process of the experiment, what would you do differently in order to find where your yield decreased?

Be sure to also answer the discussion questions regarding “Problems and Errors” and “Learning” as indicated in your lab notebook handout.